```java
import java.util.Arrays;

public class VectorOperations
{
    private double[] v1;

    // constructor
    public VectorOperations(double[] v) throws CheckDimensionsException
    {
        if(v==null || v.length==0)
            throw new CheckDimensionsException("invalid vector components");

        if(v.length!=2 && v.length!=3)
            throw new CheckDimensionsException("only 2D and 3D vectors are
allowed");

        v1 = v.clone();
    }

    // method to check dimension compatibility
    private void checkDimensions(VectorOperations other) throws
CheckDimensionsException
    {
        if(this.v1.length != other.v1.length)
            throw new CheckDimensionsException("vectors must have same
dimensions.");
    }

    // add vectors
    public VectorOperations add(VectorOperations other) throws
CheckDimensionsException
    {
        checkDimensions(other);

        double[] sum = new double[v1.length];
        for(int i=0; i<v1.length; i++)
            sum[i] = this.v1[i] + other.v1[i];
        return new VectorOperations(sum);
    }

    // subtract vectors
    public VectorOperations subtract(VectorOperations other) throws
CheckDimensionsException
    {
        checkDimensions(other);

        double[] diff = new double[v1.length];
        for(int i=0; i<v1.length; i++)
```

```java
            diff[i] = this.v1[i] - other.v1[i];
        return new VectorOperations(diff);
    }

    // dot product
    public double dotProduct(VectorOperations other) throws
CheckDimensionsException
    {
        checkDimensions(other);

        double dot = 0;
        for(int i=0; i<v1.length; i++)
            dot += this.v1[i] * other.v1[i];
        return dot;
    }

    // print vector
    public void printVector()
    {
        System.out.println(Arrays.toString(v1));
    }
}

public class CheckDimensionsException extends Exception
{
    public CheckDimensionsException(String msg)
    {
        super(msg);
    }
}

import java.util.InputMismatchException;
import java.util.Scanner;

public class MainForVectors
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        try
        {
            System.out.print("enter dimension (2/3):");
            int length = sc.nextInt();

            double arr1[] = new double[length];
            double arr2[] = new double[length];
```

```java
            System.out.println("enter elements of vector 1:");
            for(int i=0; i<length; i++)
                arr1[i] = sc.nextDouble();

            System.out.println("enter elements of vector 2:");
            for(int i=0; i<length; i++)
                arr2[i] = sc.nextDouble();

            VectorOperations v1 = new VectorOperations(arr1);
            VectorOperations v2 = new VectorOperations(arr2);

            VectorOperations sum = v1.add(v2);
            VectorOperations diff = v1.subtract(v2);
            double dot = v1.dotProduct(v2);

            System.out.println("Addition:");
            sum.printVector();

            System.out.println("Difference");
            diff.printVector();

            System.out.println("Dot Product:" + dot);
        }

        catch(InputMismatchException e)
        {
            System.out.println("invalid numeric input");
        }
        catch(CheckDimensionsException e)
        {
            System.out.println("Error:" + e.getMessage());
        }

        finally
        {
            sc.close();
        }
    }
}
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\hp\Downloads\Programming in Java> cd "c:\Users\hp\Downloads\Programming in Java\" ; if ($?) {
javac MainForVectors.java } ; if ($?) { java MainForVectors }
enter dimension (2/3):2
enter elements of vector 1:
3 4
enter elements of vector 2:
1 2
Addition:
[4.0, 6.0]
Difference
[2.0, 2.0]
Dot Product:11.0
PS C:\Users\hp\Downloads\Programming in Java>
```

Code
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